# Lecture 20

## # Applications of determinants

→ Formula for $A^{-1}$

eg $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \dfrac{1}{\underbrace{ad - bc}_{\det A}} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

$\Rightarrow \quad A^{-1} = \dfrac{1}{\det A} C^T \rightarrow$ matrix of cofactors

→ Proof attempt

- $A A^{-1} = I$

$\det A \cdot \det A^{-1} = I$

$$\boxed{\det A^{-1}} = \frac{1}{\det A}$$

$\det A^{-1}$ expressed as $A^{-1}$, how?

$$\det A^{-1} = \bar{a}_{11} \bar{C}_{11} - \bar{a}_{12} \bar{C}_{12} \; - - - - \; (-1)^{1+n} \bar{a}_{1n} \bar{C}_{1n}$$

Dont think we are getting anywhere now, lets try the other way.

$$A^{-1} = \frac{1}{\det A} C^{T}$$

$\hookrightarrow \; a_{11} C_{11} - a_{12} C_{12} - - + (-1)^{1+n} a_{1n} C_{1n}$

$$\det A^{-1} = \det \left( \frac{1}{\det A} C^{T} \right)$$

$$\det A^{-1} = \left( \frac{1}{\det A} \right)^{n} \det C^{T}$$

$\det A^{-1} = \left(\frac{1}{\det A}\right)^n \det C^T$ ↳ how to check this.

Bon, lets see in lecture now.

$\Rightarrow \quad A^{-1} = \frac{1}{\det A} C^T$ → each element product of $(n-1)^2$ entries.

$\det A$ ↓ product of $(n)^2$ entries

$\Rightarrow$ check $\quad AC^T = \det A \, I$

$AC^T \Rightarrow (AC^T)_{11} = a_{11}C_{11} + a_{12}C_{12} \dots$
$= \det A$

$(AC^T)_{12} = a_{11}C_{21} + a_{12}C_{22} \dots$

$a_{11} \; a_{12} \; \dots \; a_{1n}$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & \cdots \\ \cdots & & \end{bmatrix}$$

$$C = \begin{bmatrix} C_{11} & C_{12} & C_{13} & \cdots \\ C_{21} & C_{22} & C_{23} & \cdots \\ \vdots & & & \\ \vdots & & & \end{bmatrix}$$

$$C^T = \begin{bmatrix} C_{11} & C_{21} & C_{31} \\ C_{12} & C_{22} & \cdots \\ C_{13} & C_{23} & \\ \vdots & & \end{bmatrix}$$

$$(AC^T)_{12} = a_{11}C_{21} + a_{12}C_{22} + a_{13}C_{23}$$

Well all the diagonals would be of the form $\sum_{i=1}^{n} a_{ki} C_{ki} = \det A$

⇒ The idea is that if we multiply any row with cofactors from a different row, then they give 0 ! ! !

⇒ Reason / Proof → The reason is because this operation is similar to taking determinant of a matrix with 2 same rows ! !

So
$$AC^T = \begin{bmatrix} \det A & & 0 \\ & \det A \; \ddots & \\ 0 & & \det A \end{bmatrix}$$

$$AC^T = \det A \cdot I$$

Hence,
$$\boxed{A^{-1} = \frac{1}{\det A} C^T}$$

$$\Rightarrow \quad Ax = b$$
$$x = A^{-1} b$$

$$x = \frac{1}{\det A} C^T b$$

$$x = \frac{C^T b}{\det A}$$

## Cramer's Rule

$$x_1 = \frac{C_{11} b_1 + C_{21} b_2 \ldots}{\det A} \rightsquigarrow \frac{\det B_1}{\det A}$$

this can be seen as determinat of A when first column is replaced by b.

$$x_2 = \frac{C_{12} b_1 + C_{22} b_2 - C_{n2} b_n}{\det A} = \frac{\det B_2}{\det A}$$

Second column replaced.

$$B_1 = \begin{bmatrix} b_1 & a_{12} & \cdots \\ b_2 & a_{22} & \\ \vdots & a_{23} & \cdots \\ b_n & \vdots & . \end{bmatrix}$$

$$x_j = \frac{\det B_j}{\det A}$$

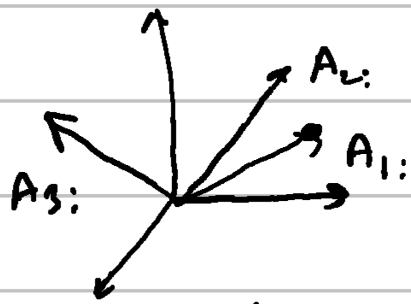
$\Rightarrow$ Cramer's rule is very bad in practice in terms of computation cost.

$\Rightarrow$ <u>Volume</u>

# determinant of a matrix is a volume of a box.

$|\det A| =$ volume of a box

* $A_{3\times 3} \rightarrow$




three corners of a "box"

$\Rightarrow$ The sign of the determinant tells us about "handedness" of a box.

eg $\boxed{A} = I$ | $\det A = I$ || $\underline{\underline{\det A = 1}}$
$\hookrightarrow$ cube

eg Orthogonal matrices $\rightarrow$

$A = Q$ then $\underline{\boxed{\det A = 1}}$ !!

a cube as well just varying orientation

$Q^T Q = I$

$\det(Q^T)\ \det(Q) = 1$

$\det(Q^T) = \dfrac{1}{\det Q}$

$\Rightarrow \quad (\det Q)^2 = 1$

$\boxed{\det Q = \pm 1}$

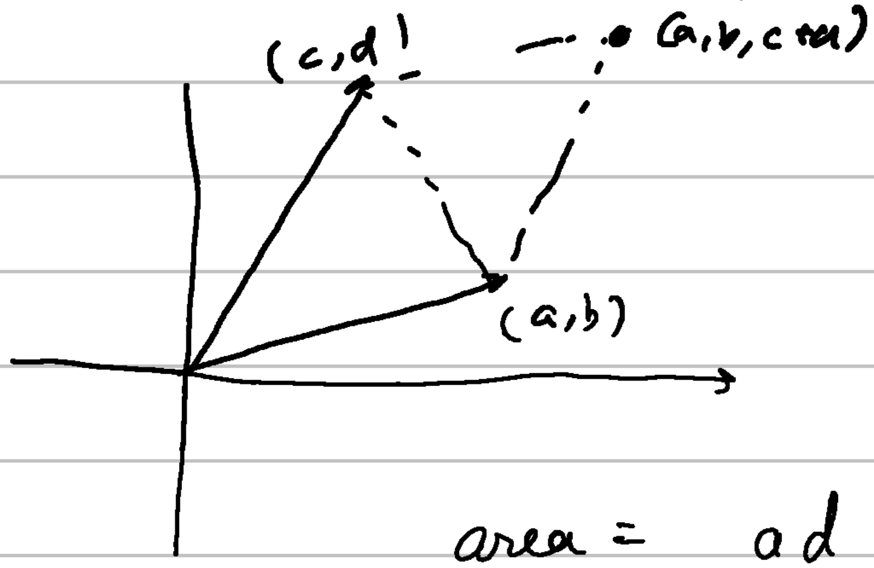
eg Rectangular cases

Lets check the 90° edges for now.

- If we double an edge, the volume doubles.

$\Rightarrow$ Similarly if we double a row, the determinant doubles!!

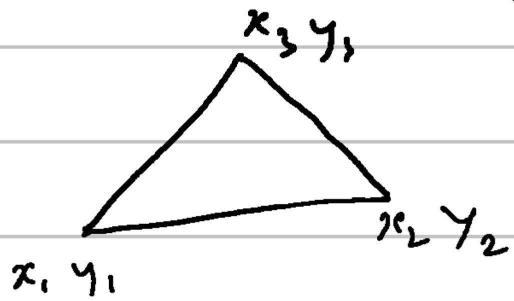
eg $$\begin{vmatrix} a+a' & b+b' \\ c & d \end{vmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} + \begin{vmatrix} a' & b' \\ c & d \end{vmatrix}$$




$$\text{area} = ad - bc$$

$$\Delta\ \text{area} = \frac{1}{2}(ad - bc)$$

triangle




$$\text{area} = \frac{1}{2}\left( \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} \right)$$